Eco-Evolutionary Forecasting of Albatross

MATLAB code repository

This repository contains the MATLAB code used to generate all eco-evolutionary simulations and figures in the accompanying manuscript:

Climate change outpaces evolutionary adaptation in a long-lived seabird population by Jenouvrier et al. submitted.

The code implements a **stage-structured, evolutionarily explicit population model** that propagates uncertainty from **climate forcing → breeding values →phenotypes → vital rates → population dynamics**, and evaluates the limits of evolutionary rescue under historical and future climate scenarios.

1. Model overview

The model tracks a population structured by:

- **Breeding value (genetic component)**
- **Phenotype (expressed trait)**
- **Age / breeding stage**

Evolution follows standard quantitative-genetic principles, with heritability controlling the partitioning of phenotypic variance into additive genetic and environmental components. Phenotypes influence survival and reproduction through empirically estimated vital-rate functions.

Each simulation combines:

- a **focal phenotype**
- a **heritability value**
- a **climate scenario**
- (for some figures) an alternative **selective-pressure scenario**

2. Phenotypes encoded in the simulations

Each simulation is run for **one phenotype at a time**, selected using the variable `PHENOTYPE`:

PHENOTYPE	Trait	Trait category
1	Activity (number of landings / take-offs)	Foraging behavior
2	Time on water	Foraging behavior
3	Return date	Phenological trait
4	Wing length	Morphological trait

3. Heritability values

Each phenotype is simulated across a range of **heritability values between 0 and 1.**

4. Climate scenarios

Climate forcing is controlled by the variable `caseID`:

caseID	Scenario	Description
1	High emissions	SSP3-7.0
3	Low emissions	Paris-compatible 2 °C pathway

Historical simulations use observed environmental conditions and do **not** vary climate through time.

5. Repository structure

Repository structure (organized by manuscript figures)

The repository is organized **by manuscript figure**, with each figure contained in its own folder. Each figure folder includes:

1. A **main figure script** (`MainFigX.m`) that reproduces the figure.
2. A **model-building script** (`main_Pheno_ENV_*.m`) specific to the environmental or selective scenario.
3. **Parameter functions** defining phenotype–environment–vital-rate relationships.
4. **Common core model functions** shared across all figures.
5. **Empirical data files** used to parameterize the model.

Figure 2 — Historical environment (no temporal climate change)

📁 `Figure2/`

This folder contains simulations under **historical environmental conditions**, with climate fixed through time.

Figure-specific files

```
MainFig2.m
main_Pheno_ENV_HIST.m
parameterPHENOTYPE_HIST.m
figure2.pdf
resultsMATfiles.zip
```

- `MainFig2.m`
  Main script that loads simulation outputs and generates Figure 2.

- `main_Pheno_ENV_HIST.m`
  Builds and runs the eco-evolutionary model under **historical (stationary) environmental conditions**.
- `parameterPHENOTYPE_HIST.m`
  Defines phenotype–vital rate relationships for the historical environment.

Figure 3 — Future climate projections with uncertainty

📁 `Figure3/`

This folder contains simulations under **future climate change**, accounting for uncertainty in:

- climate projections,
- demographic processes.

Figure-specific files

```
MainFig3.m
main_Pheno_ENV_ClimateChange_UNC.m
parameterPHENOTYPE_ENV.m
parameterPHENOTYPE_ENVSTO.m
ClimateScenario_Case1.mat
ClimateScenario_Case2.mat
ClimateScenario_Case3.mat
ClimateScenario_Case4.mat
```

- `MainFig3.m`
  Generates Figure 3 from ensemble simulation outputs.
- `main_Pheno_ENV_ClimateChange_UNC.m`
  Core model script for **future climate projections with uncertainties**.
- `parameterPHENOTYPE_ENV.m`
  Mean phenotype–environment–vital-rate relationships.
- `parameterPHENOTYPE_ENVSTO.m`
  Stochastic version used to propagate uncertainty in climate-traits-vital rates relationships.
- `ClimateScenario_Case*.mat`
  Climate forcing data for different emission scenarios (e.g. 1=SSP3-7.0, 3=Paris 2 °C).

Figure 4 — Modified selective pressure on juvenile traits

📁 `Figure4/`

This folder explores a **counterfactual selective-pressure scenario**, modifying selection on **juvenile wing length and juvenile survival**.

Figure-specific files

```
MainFig4.m
main_Pheno_ENV_ClimateChange_meanEns.m
parameterPHENOTYPE_ENV_SP.m
figure4.pdf
ecoEvoResults_*.mat
```

- `MainFig4.m`
  Generates Figure 4.
- `main_Pheno_ENV_ClimateChange_meanEns.m`
  Model script using **mean climate ensembles** with altered selective pressure.
- `parameterPHENOTYPE_ENV_SP.m`
  Defines **modified selection functions** for juvenile wing length and survival.
- `ecoEvoResults_*.mat`
  Precomputed simulation outputs used for plotting.

Files common to all figures

The following files appear in **all figure folders** and define the core eco-evolutionary model:

Core model functions

```
Umat2.m
Fmat2.m
invlogit.m
invlogitG.m
```

- `Umat2.m`
  Constructs survival and state-transition matrices.
- `Fmat2.m`
  Constructs fertility / offspring-production matrices.
- `invlogit.m`, `invlogitG.m`
  Logit functions for transforming linear predictors into probabilities.

Empirical data files

```
FOR.mat
wing_corrected.mat
Nobs_FRA.mat
```

- `FOR.mat`
  Empirical foraging behavior data.
- `wing_corrected.mat`
  Empirical wing-length measurements.
- `Nobs_FRA.mat`
  Observed population size data used for initialization and validation.

How figures are generated (summary)

For each figure:

1. `MainFigX.m`
   → sets phenotype, heritability, climate scenario, and plotting options.
2. Calls a `main_Pheno_ENV_*.m` script
   → builds the population model, runs simulations, saves results.
3. Computes derived quantities
   → population size, trait means, variances, adaptation rates.
4. Produces the final manuscript figure.

6. Program architecture (all main scripts)

Each mainFig program follows the **same three-block structure**:

Block 1 — Parameter initialization

- Load empirical trait and climate data
- Set phenotype, heritability, and climate scenario

Block 2 — Model construction and simulation

- Discretize breeding values and phenotypes
- Compute genetic and environmental variances
- Build survival (`U`) and reproduction (`F`) matrices from empirical relationships
- Construct inheritance and phenotype-mapping operators
- Run:
  - burn-in phase
  - forward eco-evolutionary projection
- Save results for **each combination of phenotype** × **climate** × **heritability** × **scenario**

Block 3 — Derived outputs for figures and make the figures

- Total population size through time
- Mean phenotype and breeding value
- Phenotypic and genetic variances
- Adaptation rates
- Phenotype-weighted vital rates

These outputs are used **directly** by the plotting scripts in each figure folder.

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